

8(a)(i)	movement of star causes change in (observed) frequency or movement of star causes redshift	B1
	observed frequency is lower (than emitted frequency)	B1
8(a)(ii)	all three lines shown to left of corresponding printed lines	B1
	distance between drawn line and corresponding printed line approximately the same for all three lines	B1
8(b)(i)	$E = hf$ and $\lambda = c / f$	C1
	$E = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (488 \times 10^{-9})$ $= 4.08 \times 10^{-19} \text{ J}$	A1
8(b)(ii)	photon energy = $(4.08 \times 10^{-19}) / (1.60 \times 10^{-19})$ $= 2.55 \text{ eV}$	C1
	energy level = $-3.40 + 2.55$ $= -0.85 \text{ eV}$	A1
8(b)(iii)	$\Delta\lambda = \lambda \times (v / c)$ $= (488 \times 6.2 \times 10^6) / (3.00 \times 10^8)$ (= 10 nm)	C1
	observed wavelength = $488 + \Delta\lambda = 488 + 10$ $= 498 \text{ nm}$	A1

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Question	Answer	Marks
8(c)	$v = H_0 d$	C1
	$d = (6.2 \times 10^6) / (2.3 \times 10^{-18})$ $= 2.7 \times 10^{24} \text{ m}$	A1